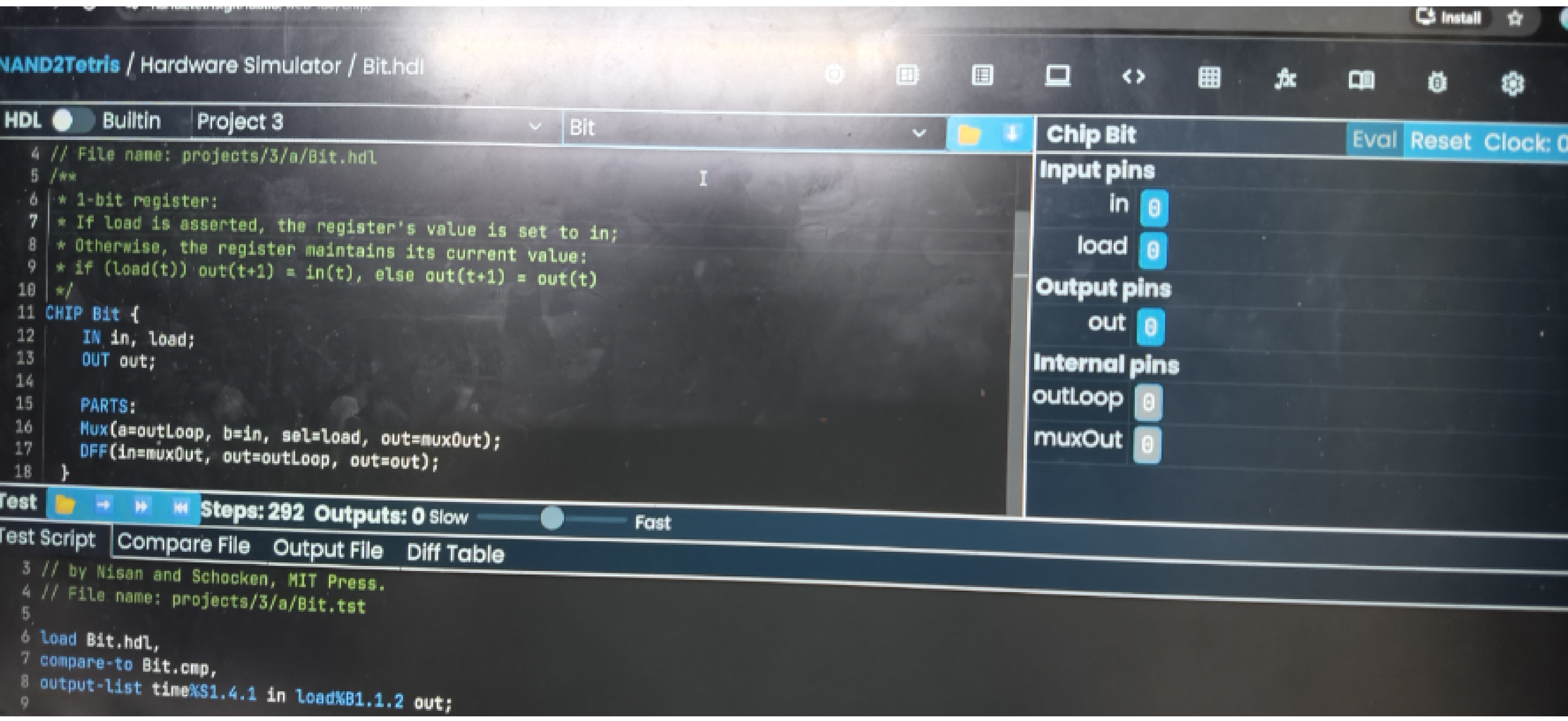

Install
NAND2Tetris / Hardware Simulator / Bit.hdl
HDL
Builtin
Project 3
Bit
// File name: projects/3/a/Bit.hdl
/**
* 1-bit register:
* If load is asserted, the register's value is set to in;
* Otherwise, the register maintains its current value:
* if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
*/
CHIP Bit {
IN in, load;
OUT out;
PARTS:
Mux(a=outLoop, b=in, sel=load, out=muxOut);
DFF(in=muxOut, out=outLoop, out=out);
}
Chip Bit
Eval
Reset
Clock: 0
Input pins
in 0
load 0
Output pins
out 0
Internal pins
outLoop 0
muxOut 0
Test
Steps: 292 Outputs: 0
Slow
Fast
Test Script
Compare File
Output File
Diff Table
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Bit.tst
load Bit.hdl,
compare-to Bit.cmp,
output-list time%S1.4.1 in load%B1.1.2 out;

HDL Builtin Project 3 Register

```
/**
CHIP Register {
    IN in[16], load;
    OUT out[16];

    PARTS:
    Bit(in=in[0] , load=load , out=out[0] );
    Bit(in=in[1] , load=load , out=out[1] );
    Bit(in=in[2] , load=load , out=out[2] );
    Bit(in=in[3] , load=load , out=out[3] );
    Bit(in=in[4] , load=load , out=out[4] );
    Bit(in=in[5] , load=load , out=out[5] );
    Bit(in=in[6] , load=load , out=out[6] );
    Bit(in=in[7] , load=load , out=out[7] );
    Bit(in=in[8] , load=load , out=out[8] );
```

Chip Register

Input pins

in 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec

load 1

Output pins

out 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec

Test Steps: 151 Outputs: 149 Slow Fast

Test Script | Compare File | Output File | Diff Table

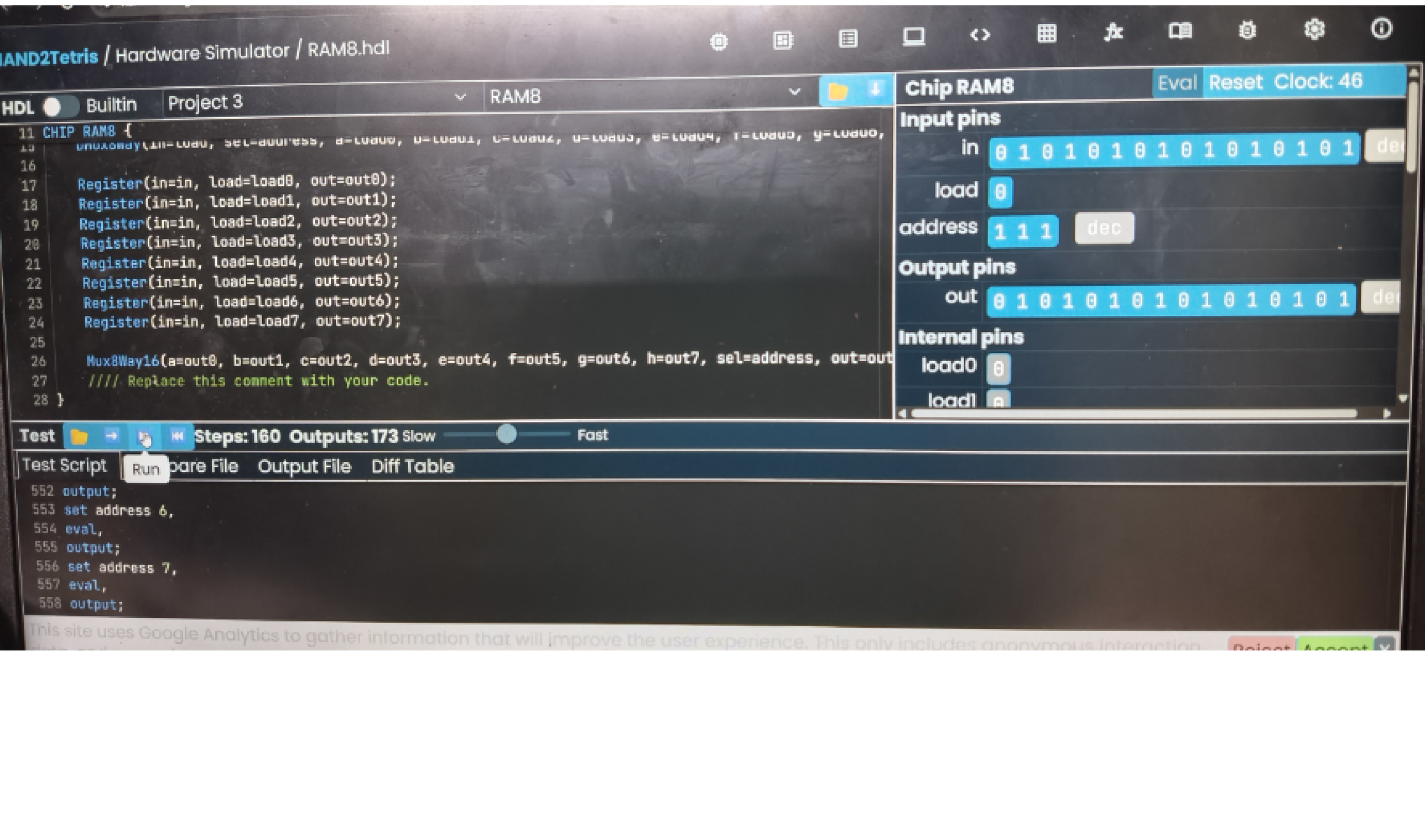
```
output;

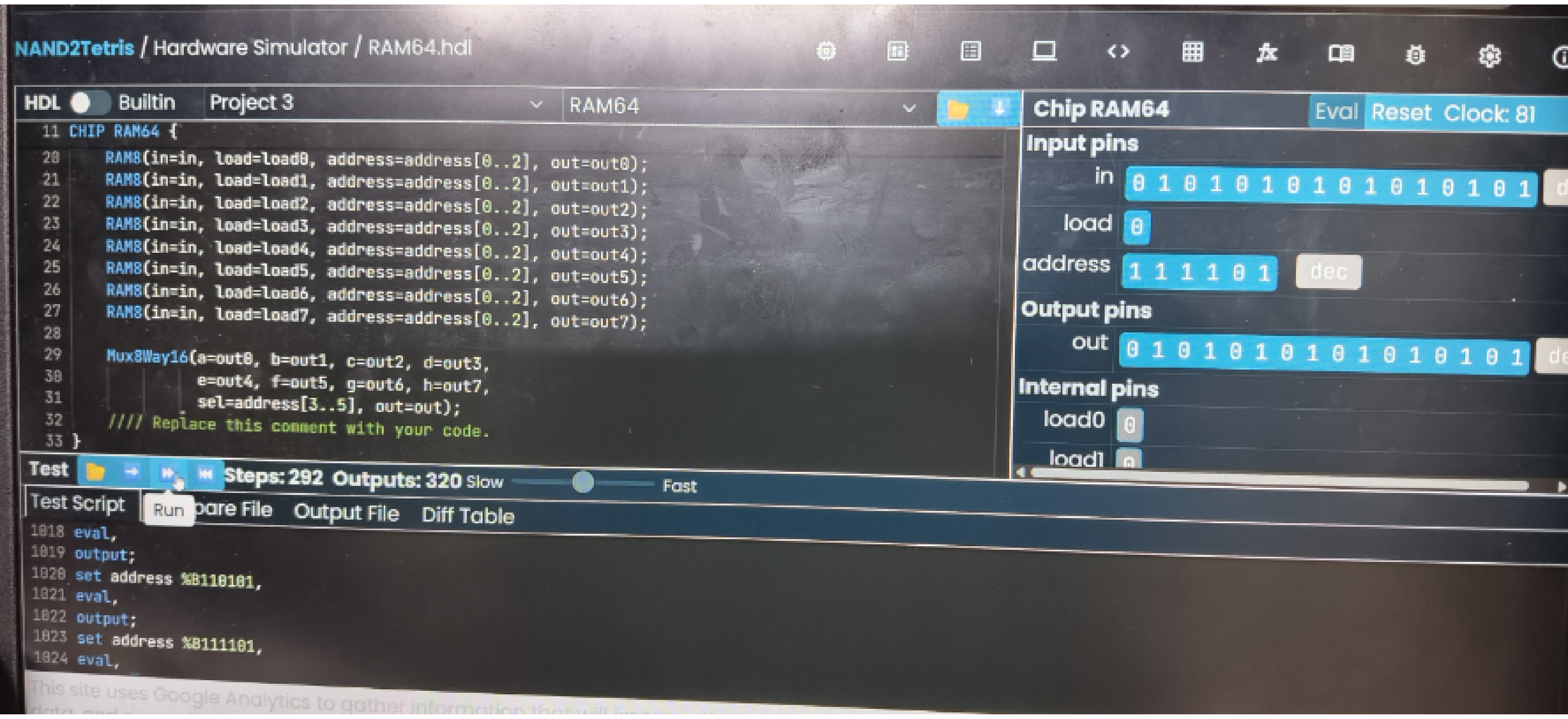
set load 1,
tick,
output;

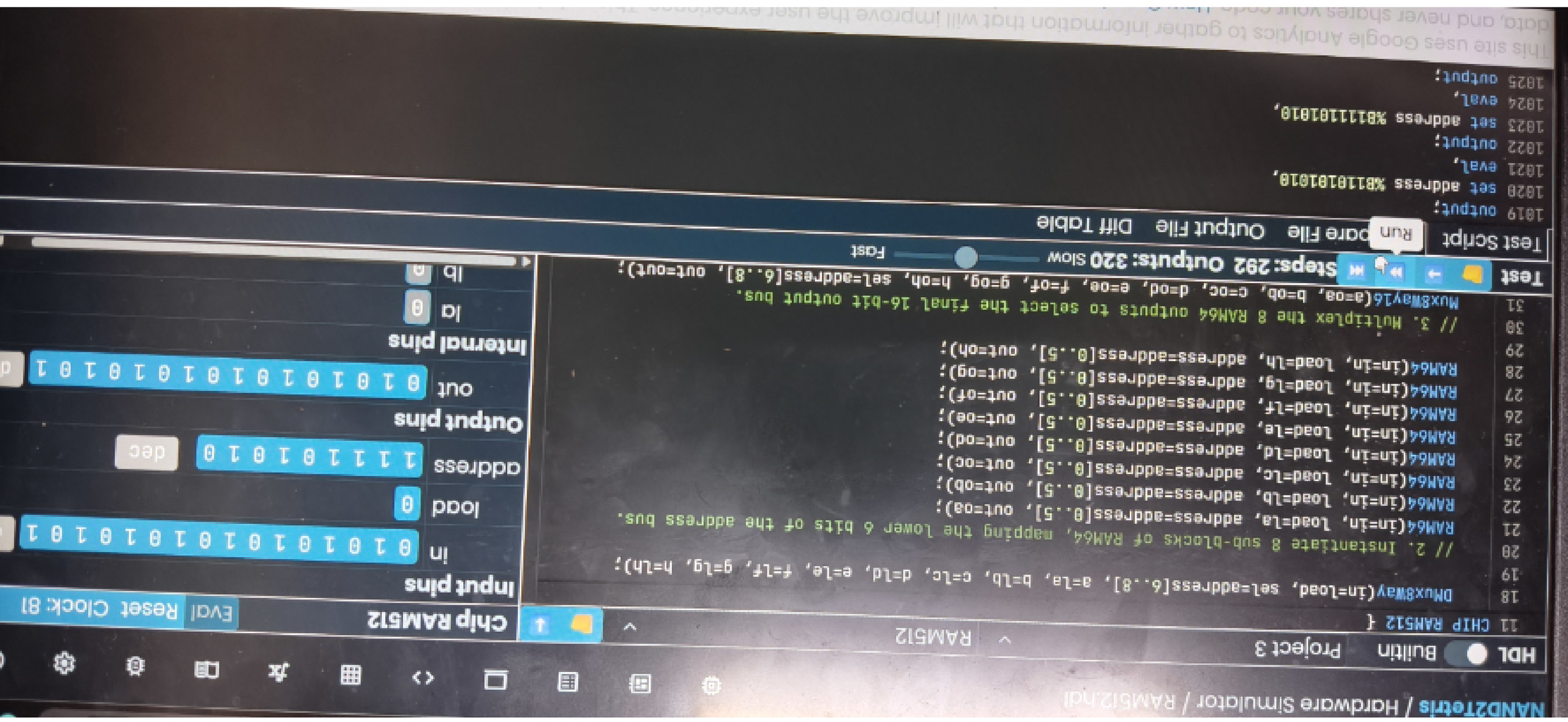
tock,
```

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Reject Accept


AND2Tetris / Hardware Simulator / RAM8.hdl
HDL
Builtin
Project 3
RAM8
11 CHIP RAM8 {
16
17 Register(in=in, load=load0, out=out0);
18 Register(in=in, load=load1, out=out1);
19 Register(in=in, load=load2, out=out2);
20 Register(in=in, load=load3, out=out3);
21 Register(in=in, load=load4, out=out4);
22 Register(in=in, load=load5, out=out5);
23 Register(in=in, load=load6, out=out6);
24 Register(in=in, load=load7, out=out7);
25
26 Mux8Way16(a=out0, b=out1, c=out2, d=out3, e=out4, f=out5, g=out6, h=out7, sel=address, out=out
27 //// Replace this comment with your code.
28 }
Chip RAM8
Eval
Reset
Clock: 46
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
load0
0
load1
Test
Steps: 160 Outputs: 173
Slow
Fast
Test Script
Run
Output File
Diff Table
552 output;
553 set address 6,
554 eval,
555 output;
556 set address 7,
557 eval,
558 output;
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NAND2Tetris / Hardware Simulator / RAM64.hdl
HDL Builtin
Project 3
RAM64
11 CHIP RAM64 {
20 RAM8(in=in, load=load0, address=address[0..2], out=out0);
21 RAM8(in=in, load=load1, address=address[0..2], out=out1);
22 RAM8(in=in, load=load2, address=address[0..2], out=out2);
23 RAM8(in=in, load=load3, address=address[0..2], out=out3);
24 RAM8(in=in, load=load4, address=address[0..2], out=out4);
25 RAM8(in=in, load=load5, address=address[0..2], out=out5);
26 RAM8(in=in, load=load6, address=address[0..2], out=out6);
27 RAM8(in=in, load=load7, address=address[0..2], out=out7);
28
29 Mux8Way16(a=out0, b=out1, c=out2, d=out3,
30 e=out4, f=out5, g=out6, h=out7,
31 sel=address[3..5], out=out);
32 //// Replace this comment with your code.
33 }
Chip RAM64
Eval
Reset
Clock: 81
Input pins
in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load 0
address 1 1 1 1 0 1
dec
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
load0 0
load1 0
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Run
Output File
Diff Table
1018 eval,
1019 output;
1020 set address %B110101,
1021 eval,
1022 output;
1023 set address %B111101,
1024 eval,
This site uses Google Analytics to gather information


NAND2Tetris / Hardware Simulator / RAM512.hdl
HDL
Builtin
Project 3
RAM512
Chip RAM512
Eval
Reset
Clock: 81
Input pins
in
load
address
1 1 1 1 0 1 0 1 0
dec
Output pins
out
Internal pins
ia
0
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare file
Output file
Diff table
Run
11 CHIP RAM512 {
18 DMux8Way(in=load, sel=address[6..8], a=la, b=lb, c=lc, d=ld, e=le, f=lf, g=lg, h=lh);
19
20 // 2. Instantiate 8 sub-blocks of RAM64, mapping the lower 6 bits of the address bus.
21 RAM64(in=in, load=la, address=address[0..5], out=oa);
22 RAM64(in=in, load=lb, address=address[0..5], out=ob);
23 RAM64(in=in, load=lc, address=address[0..5], out=oc);
24 RAM64(in=in, load=ld, address=address[0..5], out=od);
25 RAM64(in=in, load=le, address=address[0..5], out=oe);
26 RAM64(in=in, load=lf, address=address[0..5], out=of);
27 RAM64(in=in, load=lg, address=address[0..5], out=og);
28 RAM64(in=in, load=lh, address=address[0..5], out=oh);
29
30 // 3. Multiplex the 8 RAM64 outputs to select the final 16-bit output bus.
31 Mux8Way16(a=oa, b=ob, c=oc, d=od, e=oe, f=of, g=og, h=oh, sel=address[6..8], out=out);
1819 output;
1820 set address %B1101010101010,
1821 eval,
1822 output;
1823 set address %B1111101010,
1824 eval,
1825 output;
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NAND2Tetris / Hardware Simulator / RAM4K.hdl

HDL Builtin | Project 3 | RAM4K

```
11 CHIP RAM4K {
14
15     PARTS:
16     DMux8Way(in=load, sel=address[9..11], a=loada, b=loadb, c=loadc, d=loadd, e=loade, f=loadf, g=
17     RAM512(in=in, load=loada, address=address[0..8], out=outa);
18     RAM512(in=in, load=loadb, address=address[0..8], out=outb);
19     RAM512(in=in, load=loadc, address=address[0..8], out=outc);
20     RAM512(in=in, load=loadd, address=address[0..8], out=outd);
21     RAM512(in=in, load=loade, address=address[0..8], out=oute);
22     RAM512(in=in, load=loadf, address=address[0..8], out=outf);
23     RAM512(in=in, load=loadg, address=address[0..8], out=outg);
24     RAM512(in=in, load=loadh, address=address[0..8], out=outh);
25     Mux8Way16(a=outa, b=outb, c=outc, d=outd, e=oute, f=outf, g=outg, h=outh, sel=address[9..11],
26
27     //// Replace this comment with your code.
```

Chip RAM4K | Eval | Reset | Clock: 81

Input pins

in	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load	0
address	1 1 1 1 0 1 0 1 0 1 0 1 dec

Output pins

out	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1

Internal pins

loada	0
loadb	0

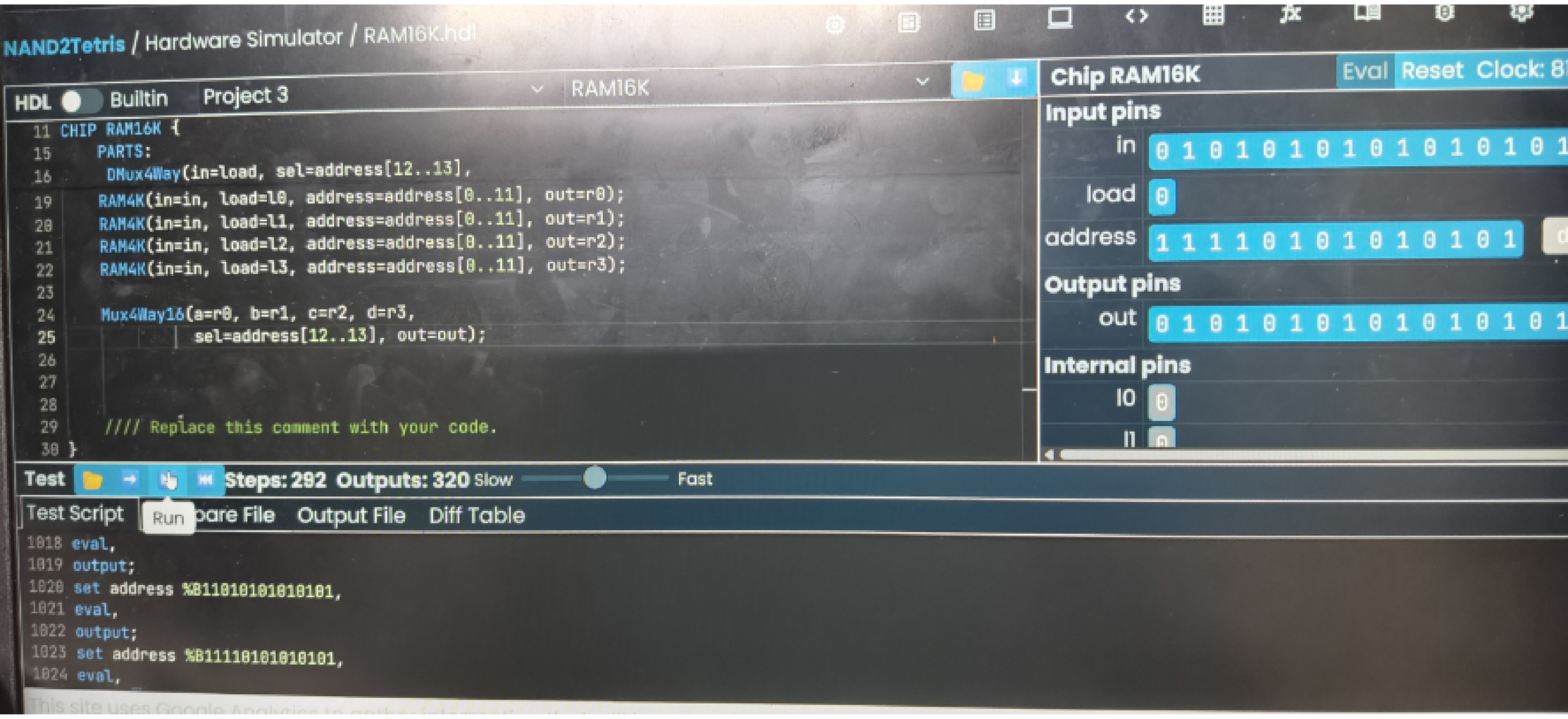
Test | Steps: 292 Outputs: 320 Slow — Fast

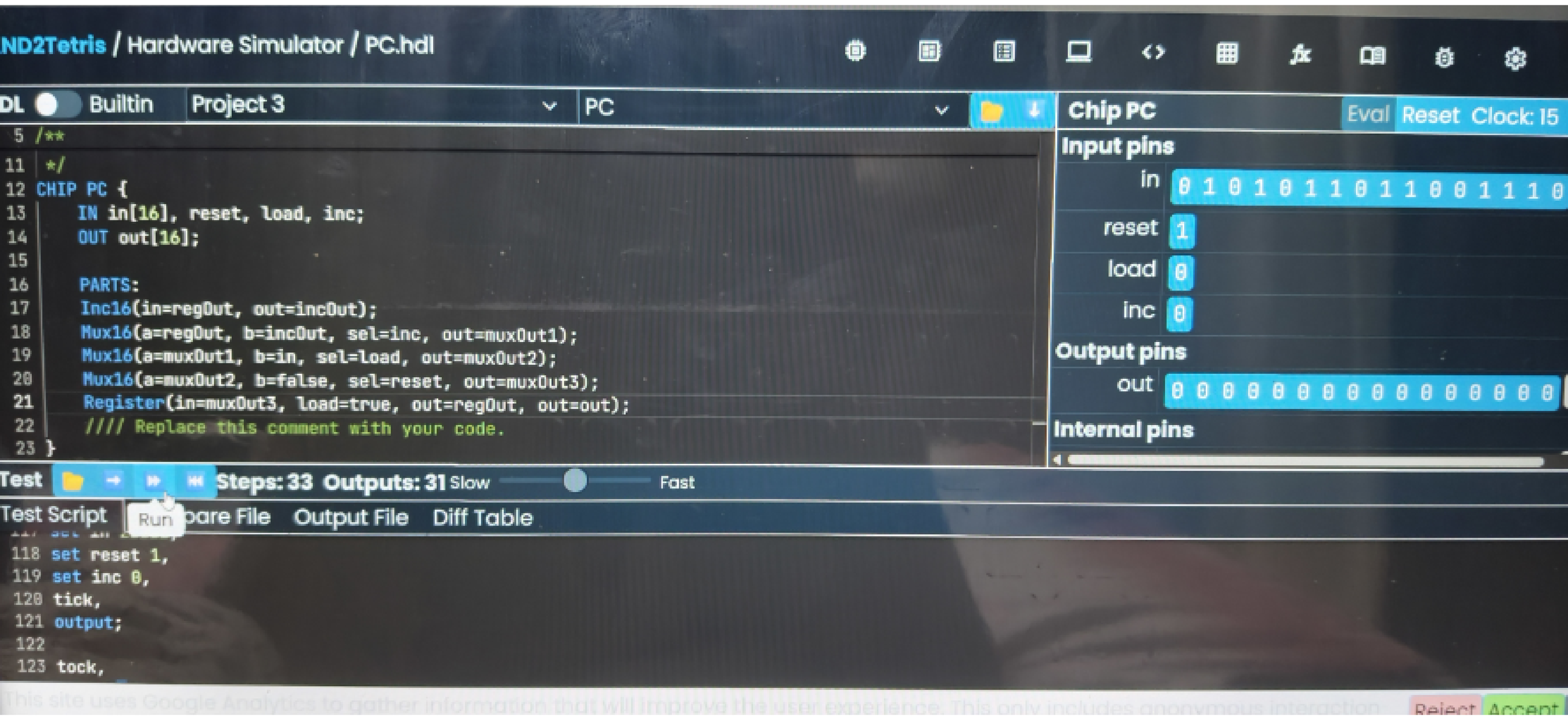
Test Script | Compare File | Output File | Diff Table

```
1018 eval,
1019 output;
1020 set address %B1101101010101,
1021 eval,
1022 output;
1023 set address %B1111101010101,
1024 eval,
```

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NAND2Tetris / Hardware Simulator / RAM16K.hdl
HDL
Builtin
Project 3
RAM16K
11 CHIP RAM16K {
15 PARTS:
16 DMux4Way(in=load, sel=address[12..13],
19 RAM4K(in=in, load=l0, address=address[0..11], out=r0);
20 RAM4K(in=in, load=l1, address=address[0..11], out=r1);
21 RAM4K(in=in, load=l2, address=address[0..11], out=r2);
22 RAM4K(in=in, load=l3, address=address[0..11], out=r3);
24 Mux4Way16(a=r0, b=r1, c=r2, d=r3,
25 sel=address[12..13], out=out);
29 //// Replace this comment with your code.
30 }
Chip RAM16K
Eval
Reset
Clock: 8
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1 0 1 0 1 0 1 0 1 0 1
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
10
0
11
0
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Run
pare File
Output File
Diff Table
1018 eval,
1019 output;
1020 set address %B1101010101010101,
1021 eval,
1022 output;
1023 set address %B1111010101010101,
1024 eval,


ND2Tetris / Hardware Simulator / PC.hdl
DL Builtin
Project 3
PC
Chip PC
Eval
Reset
Clock: 15
Input pins
in 0 1 0 1 0 1 1 0 1 1 0 0 1 1 1 0
reset 1
load 0
inc 0
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Internal pins
Test
Steps: 33 Outputs: 31
Slow
Fast
Test Script
Run
pare File
Output File
Diff Table
118 set reset 1,
119 set inc 0,
120 tick,
121 output;
122
123 tock,